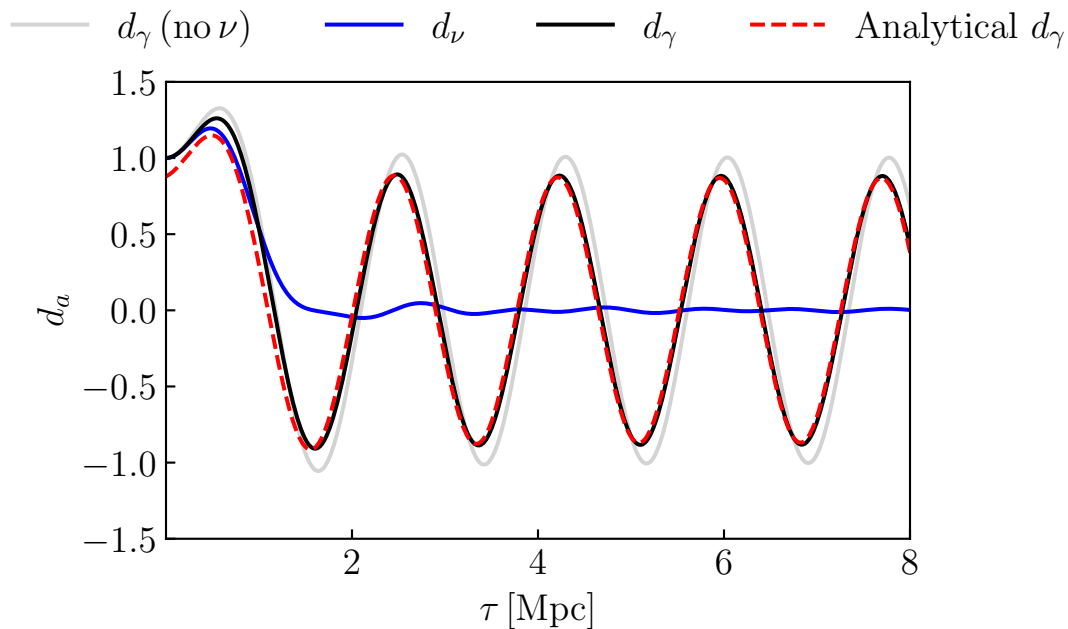


Boltzmann Code - Neutrino-Induced Phase Shift

Today we'll start learning about numerically evolving the linear perturbations. We'll start by considering a simpler system that has some cute physics in it instead of immediately trying to solve the full Λ CDM system. In particular, we'll consider a universe that contains only photons and neutrinos. What we'll find is that the neutrinos will leave an interesting imprint on the photon perturbations that is hard (possibly impossible) to mimic with any other standard-model physics: a phase shift. At the end of this lab you will have numerically solved the fully relativistic evolution of the neutrino, photon, and metric perturbations and recreate the following plot that illustrates the ν -induced phase shift:



For more detailed physical discussion see [lecture_notes/section05.pdf](#)

Assignment: We'll work in conformal Newtonian gauge with metric

$$ds^2 = a^2(\tau) [-(1 + 2\Phi) d\tau^2 + (1 - 2\Psi) d\vec{r}^2],$$

For our radiation-only universe with photons and (massless) neutrinos,

$$\rho_{\text{tot}}(a) = \frac{\rho_{\gamma,0} + \rho_{\nu,0}}{a^4}, \quad R_\nu \equiv \frac{\rho_{\nu,0}}{\rho_{\gamma,0} + \rho_{\nu,0}},$$

and we'll set $R_\nu = 1/2$ (in the real world, $R_\nu \approx 0.41$). Throughout this lab, we'll focus on a single Fourier mode with wave-number $k = 2\pi \text{ Mpc}^{-1}$.

We'll follow [Bashinsky & Seljak \(2003\)](#) and simplify our life by evolving the perturbation variable

$$d_a \equiv \frac{\delta n_{a,\text{coo}}}{n_{a,\text{coo}}} = \frac{\delta_a}{1 + w_a} - 3\Psi$$

where δ_a is the usual $\delta\rho_a/\rho_a$. For radiation ($w_a = 1/3$) this is just $d_a = (3/4)\delta_a - 3\Psi$. We also need a velocity potential u_a (so $v_{i,a} = -\nabla_i u_a$) and an anisotropic-stress potential π_a .

To fully characterize the neutrino phase-space distribution we need the higher- l multipoles $d_{l,\nu}$ for $l = 0, 1, 2, \dots$ where

$$d_{0,\nu} = d_\nu, \quad d_{1,\nu} = u_\nu, \quad d_{2,\nu} = \frac{3}{2}\pi_\nu,$$

and the higher l are independent state variables. For physical intuition for what these multipole moments are, see [lecture_notes/section05.pdf](#).

A nice thing about working with d_a instead of δ_a is that the metric perturbations Φ and Ψ are determined algebraically at every time step. We don't need to write down a e.g. $d\Phi/d\tau$ and evolve the metric perturbations as well. In particular

we have

$$\Psi = -\frac{\gamma_{\text{tot}} (d_{\text{tot}} + 3\mathcal{H} u_{\text{tot}})}{k^2 + 3\gamma_{\text{tot}}}, \quad \Phi = \Psi - 3\gamma_{\text{tot}} \pi_{\text{tot}}. \quad (1)$$

Here $\gamma_a \equiv 4\pi G a^2 (\bar{\rho}_a + \bar{P}_a)$ is the species enthalpy density, $\gamma_{\text{tot}} = \sum_a \gamma_a$, and the totals are weighted averages

$$d_{\text{tot}} = \sum_a x_a d_a, \quad u_{\text{tot}} = \sum_a x_a u_a, \quad \pi_{\text{tot}} = \sum_a x_a \pi_a, \quad x_a \equiv \frac{\gamma_a}{\gamma_{\text{tot}}}.$$

Throughout this lab, we'll use a simplified version of what we did in **background.pdf** for the full Λ CDM case

```

1 bg = {}
2
3 # Critical density for h = 1: rho_crit,100 = 3(100 km/s/Mpc)^2 / (8 pi G)
4 # for simplicity (lazyness) I'll just assume h = 1 in this program
5 _H100_SI = 100.0 * 1e3 / Mpc_in_m
6 _rho_crit_100 = 3.0 * _H100_SI**2 / (8.0 * np.pi * G)
7
8 R_nu = 0.5
9
10 # set-up with nu cosmology
11 bg['rho_gamma'] = _rho_crit_100 * (1 - R_nu)
12 bg['rho_nu'] = _rho_crit_100 * R_nu
13
14 # set-up no nu cosmology
15 bg_no_nu = {}
16 bg_no_nu['rho_gamma'] = _rho_crit_100
17 bg_no_nu['rho_nu'] = 0
18
19 def rho_total(a, bg):
20     #Total energy density at scale factor a [kg/m^3].
21     return ((bg['rho_gamma'] + bg['rho_nu']) / a**4)
22
23 def hubble(a, bg):
24     #Hubble parameter H(a) = sqrt(8 pi G / 3 * rho_total) in 1/Mpc.
25     return np.sqrt((8.0 * np.pi * G / 3.0) * rho_total(a, bg) * (Mpc_in_m / c_SI)**2 )

```

You should feel free to copy this block for `sol/nu_phase_shift.py` since you've already coded up the complete version before.

Part 1: Write `metric_perturbations(d_g, u_g, d_n, u_n, pi_n, a, aH, bg)` that returns (Φ, Ψ) using Eq. (1).

We now need to set-up the initial conditions at $\tau = \tau_0$ for our perturbations. We'll assume adiabatic initial conditions and start the integration when the mode is still frozen on superhorizon scales ($k \ll aH$). In particular we have

$$d_\gamma = d_\nu = -3\zeta, \quad u_\gamma = u_\nu = \frac{5\tau_0 \zeta}{15 + 4R_\nu}, \quad \pi_\nu = \frac{2\tau_0^2 \zeta}{3(15 + 4R_\nu)}, \quad (2)$$

with all higher- l neutrino multipoles being vanishingly small on superhorizon scales. Here ζ is the comoving curvature perturbation imprinted by inflation. We'll set $\zeta = -1/3$ in the script which makes the initial condition $d_a = +1$. Something notable: $d_a = -3\zeta$ for all species! This is a lot simpler than the ICs we'd need to set for δ_a .

Part 2: Our state-vector layout will be

$$y = \left(\underbrace{d_\gamma, u_\gamma, d_\nu, u_\nu, \pi_\nu}_{y[0..4]}, \underbrace{d_{3,\nu}, d_{4,\nu}, \dots, d_{\text{LMAX},\nu}}_{y[5..\text{LMAX}+2]} \right).$$

We can use `LMAX = 100` in this lab. Write `initial_state(R_nu)` that returns the initial condition of the full state vector using Eq. (2).

Once we have our initial conditions, all we need are the evolution equations. In the two-species universe these are

- **Photons (perfect fluid):**

$$\dot{d}_\gamma = -k^2 u_\gamma, \quad \dot{u}_\gamma = \frac{1}{3} d_\gamma + \Phi + \Psi. \quad (3)$$

- **Neutrinos, low l :** same monopole/dipole structure but with $\pi_\nu \neq 0$, plus the quadrupole

$$\dot{d}_\nu = -k^2 u_\nu, \quad \dot{u}_\nu = \frac{1}{3}d_\nu - k^2 \pi_\nu + \Phi + \Psi, \quad \dot{\pi}_\nu = \frac{4}{15}u_\nu - \frac{2}{5}k^2 d_{3,\nu}. \quad (4)$$

- **Neutrinos, higher l :** For higher- l we have the recursion ($l \geq 3$)

$$\dot{d}_{l,\nu} = \frac{l}{2l+1} d_{l-1,\nu} - \frac{l+1}{2l+1} k^2 d_{l+1,\nu}, \quad (5)$$

where at $l = 3$ you should set $d_{l-1,\nu} = d_{2,\nu} = (3/2)\pi_\nu$.

- **Streaming closure at $l = L_{\max}$:** we have to truncate the tower somewhere. From the standard [Ma & Bertschinger \(1995\)](#) streaming closure we get

$$\dot{d}_{L_{\max},\nu} = d_{L_{\max}-1,\nu} - \frac{L_{\max}+1}{\tau} d_{L_{\max},\nu}. \quad (6)$$

Part 3: Write `rhs(tau, y, bg)` that assembles the time derivatives using Eqs. (3), (4), (5), (6). You now have everything you need to integrate the system with `scipy.integrate.solve_ivp`.

Validation: You should compute $d_\gamma(\tau)$ for the case where there are no neutrinos ($R_\nu = 0$) and including neutrinos ($R_\nu = 1/2$). Plot $d_\gamma(\tau)$ for both these cases. You should see the same picture as on the first page of this handout. The two curves oscillate at the same frequency (set by the photon sound speed $c_s^2 = 1/3$). However, when ν are included, the d_γ is shifted to later times and slightly damped relative to the no- ν curve.

You could've actually analytically derived what the phase-shift and amplitude modulation should be (see Bashinsky & Seljak (2003)). In particular, we expect on sub-horizon scales

$$d_\gamma(\tau; R_\nu) \approx (1 + \Delta_\gamma) d_\gamma^{(R_\nu=0)} \left(\tau + \frac{\delta\phi \sqrt{3}}{k} \right), \quad (7)$$

where the phase shift and amplitude correction are

$$\delta\phi = 0.1912 \pi R_\nu, \quad 1 + \Delta_\gamma = 1 - 0.2683 R_\nu.$$

What you should find is that this analytical prediction should match our $R_\nu = 0.5$ pretty well throughout the subhorizon regime.

References:

- **Peebles (1973)**, *The Role of Neutrinos in the Evolution of Primeval Adiabatic Perturbations*, ApJ 180, 1.

This is the calculation we're reproducing in this lab. As far as I can tell, this is the first time the phase shift due to neutrinos was seen in the literature. However, Peebles did not say much about it in this paper, you can only really see the phase shift by reading Table. 1.

- **Bashinsky & Seljak (2003)**, *Signatures of relativistic neutrinos in CMB anisotropy and matter clustering*, [astro-ph/0310198](#).

The first paper explicitly pointing out this phase shift and building a detailed understanding of its origin. There is also a lot of discussion about how to think about perturbations in real space here. The first author (Bashinsky) spent a lot of time developing Green's function techniques to analytically understand perturbations in real space. This approach is complementary to the primarily Fourier space treatment that most textbooks take. This paper isn't the friendliest read...

- **Baumann, Green, Meyers & Wallisch (2015)**, *Phases of New Physics in the CMB*, [arXiv:1508.06342](#).

Includes a pedagogical derivation of the phase shift and how some discussion of what this means for BSM physics searches. There is also a followup by some of the same people looking at the phase shift in BAO.

- **Follin, Knox, Millea & Pan (2015)**, *First Detection of the Acoustic Oscillation Phase Shift Expected from the Cosmic Neutrino Background*, [arXiv:1503.07863](#).